Yiyang Lu

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Education

IIIS, Tsinghua University
Undergraduate Study Sep 2024 - Now

IIIS, Tsinghua University
Pre-College Program Sep 2023 - Jun 2024

Research interests

Computer Vision, Robotics, Reinforcement Learning

Experiences

Undergraduate Research in Computer Vision
Supervised by Professor Kaiming He, Massachusetts Institute of Technology Feb 2025 - Now

- Conduct research on generative models, including Diffusion models, Flow matching, and Normalizing Flows.
- Develop and deploy models on Google Cloud Platform using TPU and JAX.

Undergraduate Research in Robotics
Supervised by Professor Huazhe Xu, Tsinghua University Sep 2024 - Jun 2025

- Gain experiences on robotics and computer vision, especially designing diffusion models and vision encoder for imitation learning.
- Implementing ideas and training models on real robots and simulators, such as Metaworld, Maniskill, Robotwin.

Internship in Computer Vision
Supervised by Dong Wei, Tencent Jul 2024

- Worked on multi-modal AI, especially extracting features from images and texts.
- Selected as ”Challenging Star” in the internship program.

Publications

H^3DP: Triply-Hierarchical Diffusion Policy for Visuomotor Learning
Yiyang Lu, Yufeng Tian*, Zhecheng Yuan*, Xianbang Wang, Pu Hua, Zhengrong Xue, Huazhe Xu*

https://arxiv.org/abs/2505.07819

Awards

Tsinghua Academy Talent Development Program Scholarship
Tsinghua University 2024

Freshman Scholarship (Second Prize)
Tsinghua University 2024

Gold Medal (Top 10)
39th Chinese Physics Olympiad 2022

Projects

Flow-made Anime Characters
2025.5 https://github.com/PeppaKing8/dl_project

Project for the course *Deep Learning* at IIIS, Tsinghua University, completed in a team of 2. This project is selected as **the most popular project (top 1)**.

- Introduce a novel Normalizing Flow framework that significantly improves image generation quality by incorporating score-based denoising and a flexible, frequency-domain prior.
- Achieve state-of-the-art Fréchet Inception Distance (FID) scores on the AnimeFace dataset, generating high-fidelity, coherent images and learning a smooth, semantically meaningful latent space.
- Reveal a visual resemblance between the sampling trajectories of flow and those of diffusion models, suggesting a deeper connection between these distinct generative paradigms.

Realistic Multi-Material Interaction Simulation Pipeline
2024.12 https://github.com/Lyy-iiis/ACG-Project

Project for the course *Advanced Computer Graphics* at IIIS, Tsinghua University, completed in a team of 2. This project is selected for class **oral presentation (top 3)** and **the most popular project (top 3)**.

- Implement a comprehensive simulation pipeline that supports multiple materials and their interactions, including rigid body, fluid, cloth and smoke simulation.
- Leverages GPU acceleration and spatial hashing to achieve high simulation performance and scalability, produce real-time visual effects.

Do LLMs Outperform Multi-task Learning Expert in Medical Report Generation?
2024.12 https://github.com/Lyy-iiis/Crayon-new

Project for the course *Natural Language Processing* at IIIS, Tsinghua University, completed in a team of 3. This project is selected for class **oral presentation**.

- Compare the performance of Large Language Models and Multi-task Learning Expert in medical report generation, revealing the weaknesses of LLMs in domain-specific tasks even with fine-tuning.
- Implement a multi-task learning expert for medical report generation, which achieves competitive performance on IU-X-Ray dataset.

Randomized Techniques in Graph Algorithms
2024.10 https://github.com/PeppaKing8/algdesign-project

Project for the course *Algorithm Design* at IIIS, Tsinghua University, completed in a team of 2. This project receives **6 bonus points** for creativity.

- Investigate the application of randomized techniques in graph algorithms, as well as implementation in C++.
- Propose a new NPC problem, Optimal Point Traversing Path, and present a sublinear randomized algorithm to solve the special case, completed with a correctness proof and complexity analysis.

Music Image Transfer
2024.6 https://github.com/Lyy-iiis/LLM_project

Project for the course *Introduction to Large Language Model Application* at IIIS, Tsinghua University, completed in a team of 3. The project is further developed by **ZhiPu AI**.

- Build a pipeline of Large Language Models to generate an image according to the content of a piece of music.
- Achieve high correlation between the contents and rather high speed of generation.
- Use Docker to build an API on Kubernetes server and use gradio to build up a website for the application.

Skills

Programming Languages: Python, C/C++, Bash, Assembly, Scala
Tools: PyTorch, JAX, MATLAB, Git, LaTeX, Docker, GDB, SQL
Languages: English (Fluent), Chinese (Native)

Additional information

Transript at IIIS, Tsinghua University
All of my **professional courses** taken at IIIS, Tsinghua University is shown below.

Year-Semester	Course Title	Credit	Grade
2023-Autumn	Introduction to Computer Science	3	A+
	Introduction to Programming in C/C++	2	A+
	Calculus A (1)	5	A
	Linear Algebra	4	A+
	General Physics (2)	4	A+
2024-Spring	Introduction to Computer Systems	4	A+
	Introduction to Large Language Model Application	2	A
	Mathematics for Computer Science and Artificial Intelligence	4	A+
	Calculus A (2)	5	A+
	General Physics (1)	4	A
	Quantum Computer Science	4	A+
2024-Autumn	Advanced Computer Graphics	3	A+
	Natural Language Processing	3	A+
	Machine Learning	4	A+
	Algorithm Design	4	A+
	Artificial Intelligence: Principles and Techniques	3	A+
2025-Spring	Computer Vision	3	A
	Deep Learning	3	A
	Multi-modal Machine Learning	3	A

Self-Studied/Audited courses

- Abstract Algebra, IIIS, Tsinghua University
- The Missing Semester of Your CS Education, MIT, `https://missing.csail.mit.edu/`
- Introduction to Algorithms, MIT, `https://ocw.mit.edu/courses/6-006-introduction-to-algorithms-spring-2020/`
- Design and Analysis of Algorithms, MIT, `https://ocw.mit.edu/courses/6-046j-design-and-analysis-of-algorithms-spring-2015/`
- CS224n: Natural Language Processing with Deep Learning, Stanford, `http://web.stanford.edu/class/cs224n/`
- CS231n: Deep Learning for Computer Vision, Stanford, `http://cs231n.stanford.edu/`
- Intro to Computer Systems, CMU, `https://www.cs.cmu.edu/afs/cs/academic/class/15213-f15/www/schedule.html`
- CS285: Deep Reinforcement Learning, UC Berkeley, `https://rail.eecs.berkeley.edu/deeprlcourse`